

| Comment                | Description                                                                                           | Designator                     | Footprint                       | Quantity |
|------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------|----------|
| 100nF                  | Capacitor                                                                                             | C1, C3, C4, C11, C13, C14, C15 | C0402                           | 7        |
| 10uF/50V               | Capacitor                                                                                             | C12, C19                       | C1206                           | 2        |
| 100nF/50V              | Capacitor                                                                                             | C16                            | C0402                           | 1        |
| 22uF                   | Capacitor                                                                                             | C17, C18                       | C0603                           | 2        |
| 1uF                    | Capacitor                                                                                             | C21                            | C0603                           | 1        |
| 220uF/50V              | Polarized Capacitor (Axial)                                                                           | C24                            | CAPR-5mm                        | 1        |
| 18V                    | Zener                                                                                                 | D1                             | SOD-323                         | 1        |
| B5819W SL              | Diode                                                                                                 | D2, D6, D9                     | SOD-123                         | 3        |
| MB10S                  |                                                                                                       | D3                             | SOIC245P670X290-4N              | 1        |
| BAV99                  | Diode Array 1 Pair Series Connection Standard 100V 215mA (DC) Surface Mount TO-236-3, SC-59, SOT-23-3 | D4                             | SOT23                           | 1        |
| 3V6                    | Zener                                                                                                 | D5                             | SOD-323                         | 1        |
| 500mA                  | Fuse                                                                                                  | F1, F2                         | 1206                            | 2        |
| GZ1608D601TF           | Ferrite Bead                                                                                          | FB1                            | 0603                            | 1        |
| LG55145                | 60-V, 600-mA High Efficiency Wide Input Voltage Range Buck Converter                                  | IC1                            | SOT95P280X110-6N (SOT23-6L)     | 1        |
| C720477                | Tactile Switch                                                                                        | J2, J3                         | TS-1088-AR02016                 | 2        |
| FFC 0.5mm 10Pin 90°    | FFC 0.5mm 10P Bottom Connection C262659                                                               | J4                             | TE_1-1734592-0                  | 1        |
| Terminalblock X2       |                                                                                                       | J6, J7                         | Terminalblock_X2_5mm            | 2        |
| CMI321609X100KT (10uH) | 15uH                                                                                                  | L1                             | L1210                           | 1        |
| KF2EDGR-3.5-2Pin       | KF2EDGR-3.5-2Pin                                                                                      | P6                             | CONN-TH_2P-P3.50_KF2EDGR-3.5    | 1        |
| Header                 | 4 Pin male                                                                                            | P7                             | TE_826647-4                     | 1        |
| AO3400A                | 30V N-Channel MOSFET                                                                                  | Q1                             | SOT23 Normal                    | 1        |
| AO3401A                | 30V P-Channel MOSFET                                                                                  | Q2                             | SOT23 Normal                    | 1        |
| MMBT5401               | 60V, 600 mA, PNP switching transistor                                                                 | Q3                             | SOT23 Reverse                   | 1        |
| 2N7002                 | 30V N-Channel MOSFET                                                                                  | Q4                             | SOT23 Normal                    | 1        |
| 100k                   | Resistor                                                                                              | R1, R5, R31                    | R0603                           | 3        |
| 1                      | Resistor                                                                                              | R2                             | R0805                           | 1        |
| 1k                     | Resistor                                                                                              | R3, R29                        | 0402                            | 2        |
| 100k                   | Resistor                                                                                              | R4                             | 0402                            | 1        |
| 33k                    | Resistor                                                                                              | R6, R46                        | R0603                           | 2        |
| 10k                    | Resistor                                                                                              | R8, R9, R18, R23, R47          | 0402                            | 5        |
| 0                      | Resistor                                                                                              | R10                            | R0603                           | 1        |
| 47k                    | Resistor                                                                                              | R11                            | R0603                           | 1        |
| 15k                    | Resistor                                                                                              | R12                            | 0402                            | 1        |
| 10                     | Resistor                                                                                              | R27, R41                       | R0603                           | 2        |
| 1M                     | Resistor                                                                                              | R28                            | R0603                           | 1        |
| 4.7k                   | Resistor                                                                                              | R30                            | R0603                           | 1        |
| 8.2k                   | Resistor                                                                                              | R32                            | R0603                           | 1        |
| 18k                    | Resistor                                                                                              | R33                            | R0603                           | 1        |
| CH32V003F4U6           | CPU Core:RISC-V CPU Maximum Speed:48MHz Program Storage Size:16KB Number of I/O:18 Applications:-     | U1                             | QFN-20_L3.0-W3.0-P0.40-TL-EP1.7 | 1        |
| LMV331IDBVRG4          | Single General Purpose Low-Voltage Comparator 5-SOT-23                                                | U2                             | SOT23-5                         | 1        |
| AT24C256C-SSHL-T       | AT24C256C-SSHL-T                                                                                      | U3                             | SOIC127P600X265-8N (SOIC-8)     | 1        |